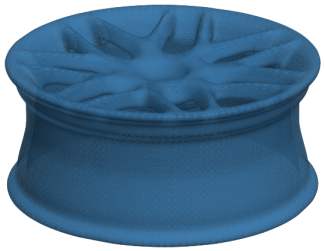
New wheel and e-bike CAD/FEM sources

12 October 2026 | Downloaded original published geometry and references

New family / source	Collected content
Wheel free-free vibration / DeepWheel	10 original STEP solid CAD models and paired FEM modal frequency labels. Full 904-row source CSV retained.
E-bike frame / TU Wien MBSE	1 native SolidWorks part with original displacement contour PDF and systems requirements. Plot-only reference.

11 new CAD/reference candidates from two sources. Ten wheel variants represent one engineering family, not ten different task families. E-bike adds a second source/design case. No geometry or FEM solve was generated locally.

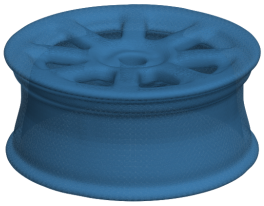


Inspection render of original epsilon_0285_2 STEP: valid single solid, 9,268 faces. The publisher reconstructed this geometry from images and converted it to BREP; it is not human-authored parametric CAD or an original dimensioned drawing.

Published scalar FEM reference	Value
Mode 7 / rim frequency	400.1808 Hz
Mode 11 / spoke frequency	1109.113 Hz

Source: DeepWheel / KAIST SmartDesignLab and Narnia Labs. Soyoung Yoo and Namwoo Kang; dataset CC BY-NC 4.0. Original files and source documentation are preserved.

Wheel modal reference table



Inspection render of original beta_0451_0 STEP. All ten imported models are valid single solids with 7,928-9,774 faces. Original filenames and checksums are indexed in the catalog.

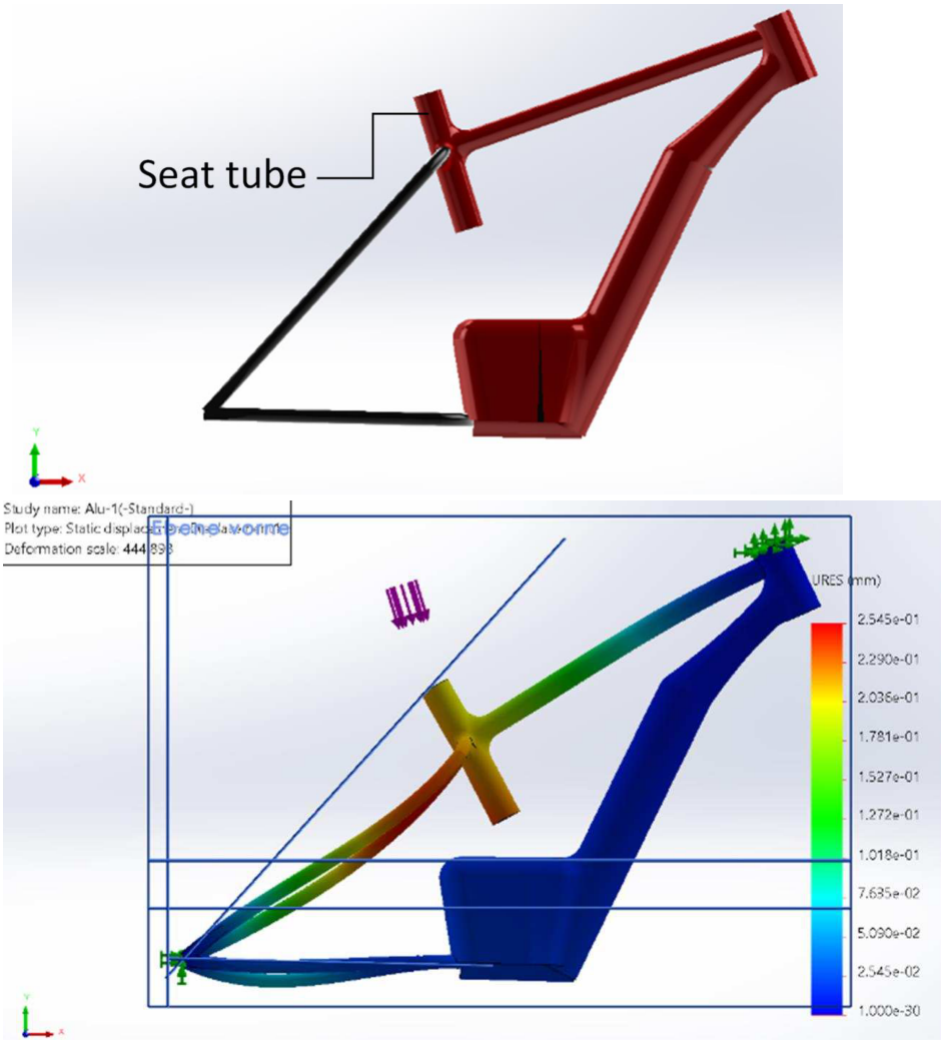
ID (full original ID in catalog)	Mass kg	Mode 7 Hz	Mode 11 Hz
W01: zeta_0133	19.7016	366.0754	916.9191
W02: zeta_0471	18.4001	390.0166	975.5839
W03: epsilon_0285_2	19.7224	400.1808	1109.113
W04: theta_0062_2	19.6203	407.125	1099.993
W05: nu_0047_3	20.2395	413.6829	1152.342
W06: theta_0042	21.3523	420.8694	1103.563
W07: beta_0446_0	20.7576	426.9859	1223.881
W08: kappa_0331	21.9739	433.5723	1319.919
W09: kappa_0581_3	22.0615	443.7821	1327.715
W10: beta_0451_0	22.9684	479.6003	1345.399

Selection: ten evenly spaced cases in sorted Mode 7 frequency order. This spans the 366.0754-479.6003 Hz source range; it is not a blind or random test split. 904 exact CSV-to-STEP filename matches verified; only ten CAD files downloaded.

Publisher setup: OptiStruct via SimLab, free-free normal modes, Tet10 volume mesh with average size 6 mm. Stated A356-T6 material: E = 72 GPa, nu = 0.33, density = 2680 kg/m3. Mode 7 is after the six rigid-body modes; detailed mode-family matching needs eigenvectors.

CONSISTENCY FLAG: CAD mass at stated density is about 3.25% below the CSV mass in every selected case. Published masses imply approximately 2770 kg/m3. This suggests a density/setup discrepancy; the actual solver input was not verified. Do not silently change density to fit labels.

E-bike frame static loading



Original publisher simulation.pdf, reproduced as an unmodified page view: frame CAD above and SolidWorks resultant displacement contour below. Native ebike-frame.SLDPRT is included.

Published contour legend	Reading
Resultant displacement URES	Displayed maximum 0.2545 mm
Reference precision	Rounded image legend, four significant digits; not a full nodal output file

Eight original source files downloaded; all publisher MD5 checksums pass. Requirements, architecture and user-needs PDFs accompany the CAD and simulation image.

Scoring gaps: force magnitude and full material/boundary setup are not readable in the supplied PDF; no raw nodal result array or neutral STEP export verified. Native geometry validity and analysis state need a SolidWorks-compatible reader. This remains a plot-reference candidate, not an automatically scored case.

Source: TU Wien Research Data, DOI 10.48436/npj9z-j2c96. Authors listed in original metadata; CC BY 4.0. Requirements diagrams do not replace dimensioned drawings.

Benchmark preparation and rejected leads

Reference category	Current use
10 wheel frequency pairs	Scalar modal prediction / supplied-CAD solver workflow candidates. Relative frequency errors can be reported after resolving setup discrepancy.
E-bike contour reference	CAD inspection and source-task audit candidate; precise automatic displacement scoring is not ready.
Drawing-to-CAD-to-FEM track	No new verified original dimensioned drawings. STEP inspection views and source pictures are not counted as engineering drawings.

Zero new certified admissions, zero model trials and zero local solver runs. Do not combine scalar frequencies with stress-field scores or treat all variants as independent task families. Public-data contamination, mesh convergence, mode ordering and replay tolerances remain to audit.

Other source crawled	Reason not added as a solved CAD/FEM pair
Composite thermal damage / Mendeley xjf5cmkjm3	Three ANSYS Workbench archives downloaded (about 100 MB). They contain DesignModeler CAD and Mechanical databases, but .skipped_files_on_archive explicitly lists the omitted .rth thermal result files and solver outputs. No solved temperature arrays verified.
Nominal tetrachiral compression / Zenodo 21669226	STEP + Abaqus CAE + experimental force-displacement records. No numerical FEM output file verified; experiments kept distinct.
Ocular adduction / Zenodo 7306040	Complex STEP assembly and Abaqus INP cases, but no published solved field/reference file verified in inspected repository.
CarHoods10k / Dryad	Structural scalars and STL geometry available; native BREP CAD not verified.
Antmicro thermal example	Native FCStd and workflow scripts found, but no saved original solved fields verified.

Package contains qualified candidate-source originals, catalogs, provenance, SHA-256 checksums, validation reports and audit scripts. The three unqualified composite archives remain in the local research cache and are excluded from the paired-source ZIP.

Earlier published CAD/FEM collections remain linked in selected-dataset.json. The new source subset is an incremental addition, not a replacement.